Executive Summary

Niyukti helps Training and Placement (T&P) teams assess applicants against job requirements. Resume screening can unfairly reward length and repeated keywords, disadvantaging concise candidates and favoring keyword-heavy resumes.

Team Atom proposes a length-robust, evidence-based scoring system. It converts each job description into fixed criteria and weights, extracts resume evidence, removes duplicates, and calculates scores with deterministic code. An LLM structures evidence but cannot change weights or award points for length or writing style.

We will test concise, expanded, keyword-stuffed, paraphrased and duplicate-evidence resume variants against a baseline. Metrics will cover score sensitivity, keyword-stuffing gain, shortlist stability, precision and recall. Uncertain cases go to human review or assessment.

WHAT SUCCESS MEANS Equivalent qualifications should produce approximately equivalent scores, while genuinely stronger job-relevant evidence may improve a score. The system must demonstrate this through controlled tests; fairness is not assumed from the formula alone.

2. Problem Statement and Motivation

2.1 The reviewer concern

How can Niyukti demonstrate that concise resumes are not disadvantaged compared with longer or keyword-heavy resumes?

Keyword frequency is not proof of capability, and LLMs may respond to length, wording or repeated claims. Resume-retrieval research also reports length and name effects, so screening systems require direct audits.

2.2 Illustrative example

Candidate A - concise resume	Candidate B - keyword-heavy resume
Lists Java, Spring Boot and SQL once, with a clear project example.	Repeats Java, Spring Boot, SQL and REST APIs across several sections.
Explains a working REST API and database integration in a few sentences.	Lists many technologies but gives limited implementation detail.
Provides relevant evidence in a short document.	Uses more words, but may provide no additional evidence.

EXPECTED BEHAVIOUR If both candidates provide equivalent evidence of equivalent qualifications, their scores should be approximately equal. Candidate B should not gain points merely by repeating a skill, and Candidate A should not lose points merely for being concise.

3. Background and Existing Approaches

Use retrieval and language technologies together, but do not treat textual relevance as proof of ability.

Approach	What it does well	Key limitation	Recommended use in Niyukti
Keyword matching	Finds exact skill terms; simple and fast.	Can miss synonyms and reward repeated words if frequency drives score.	Retrieve candidate skill mentions only; no direct points for repetition.

Approach	What it does well	Key limitation	Recommended use in Niyukti
TF-IDF	Weights terms by local frequency and collection-wide informativeness..	Still measures text relevance rather than practical proficiency; normalization does not guarantee fairness.	Optional lexical retrieval or evidence highlighting.
BM25	Ranks documents using term statistics and document-length normalization.	Designed for information retrieval; a high rank is not proof of competence or fairness.	Optional first-stage retrieval of relevant resume sections.
Semantic embeddings	Finds related meanings even when wording differs.	Similarity does not verify claims; retrieval may still reflect bias and document effects.	Locate relevant evidence and paraphrases.
LLM evaluation	Can extract skills, interpret projects and return structured information.	May vary with context and prompts and may not align consistently with expert judgments.	Evidence extraction and classification only; keep scoring in deterministic code.

3.1 Why a hybrid system is preferable

- Lexical matching finds explicit skills (e.g., Java, SQL, Git).
- Semantic retrieval finds relevant descriptions with different wording.
- Evidence classification separates claims from projects and assessment results.
- Deterministic scoring applies consistent weights and score limits.
- Counterfactual tests reveal sensitivity to irrelevant changes.

DESIGN CHOICE Use AI to locate and organize evidence; use fixed, inspectable rules to score it. Treat retrieval similarity as one signal, not the final hiring score.

4. Proposed Solution Architecture

The candidate pipeline produces an explainable recommendation; the separate validation pipeline tests score and ranking stability on controlled resume variants.

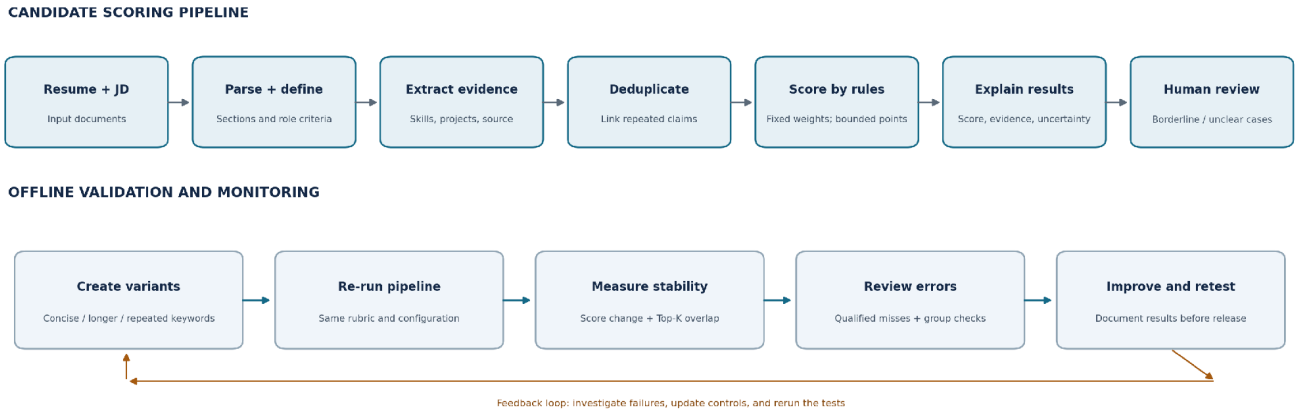


Figure 1. Proposed Niyukti scoring and validation architecture.

Component	Responsibility	Output
1. Resume and JD parser	Extract resume text, sections and job requirements.	Structured input

Component	Responsibility	Output
2. Requirement builder	Define criteria, required/preferred skills and weights for the role.	Versioned scoring rubric
3. Evidence extractor	Identify skill claims, project details, work experience and assessment evidence.	Evidence records with source location
4. Deduplication and linking	Link repeated or near-duplicate statements to the same underlying achievement.	Unique evidence items
5. Deterministic scoring engine	Apply rubric levels and fixed weights; enforce criterion caps.	Score and criterion breakdown
6. Explanation and review	Show evidence, uncertainty, missing information and review flags.	T&P recommendation
7. Bias-testing module	Run controlled resume variants and calculate evaluation metrics.	Audit report and pass/fail findings

Our Approach

NIYUKTI uses a structured, evidence-based, and bias-aware approach to evaluate resumes against job requirements. The system extracts relevant information from resumes and job descriptions, identifies verifiable skill evidence, and applies a predefined scoring rubric with fixed weights and criterion-level caps. Duplicate claims are linked to the same underlying achievement to prevent inflated scores, while missing information and uncertainty are clearly highlighted. Finally, the system generates explainable recommendations for Training & Placement (T&P) teams and conducts controlled bias testing to ensure that concise resumes are evaluated fairly alongside longer resumes with similar qualifications.

4.1 Evidence traceability

Link every criterion score to its resume excerpt, source location, evidence type, rubric level and verification status. The LLM returns structured data (e.g., JSON) but cannot change weights or invent evidence. Store assessments and reviewer overrides separately from resume claims.

5. Mathematical Mitigation Techniques

Use a fixed, job-specific rubric with bounded contributions. Cap keyword effects, deduplicate evidence and test robustness.

5.1 Fixed, job-specific weighted rubric

Illustrative rubric for a Java Backend Developer internship:

Criterion	Weight	What is assessed
Java fundamentals	25%	Evidence of core language understanding and use.
DSA and problem-solving	20%	Relevant problems, implementations or assessments.
Backend / API development	25%	API design, implementation, testing and related work.
Database knowledge	15%	Schema, queries, persistence or database integration.

Criterion	Weight	What is assessed
Projects and practical evidence	15%	Relevant implementation details and verified work.
TOTAL	100%	Weights are role-specific and should be validated.

Score each criterion from 0–5 using documented, role-specific standards agreed before evaluation. The levels shown are illustrative.

Level	Illustrative standard
0	No relevant evidence found in the supplied material.
1	Skill mentioned, but supporting evidence is absent.
2	Basic learning, coursework or simple-task evidence.
3	Relevant project or implementation evidence.
4	Strong implementation evidence with useful technical details.
5	Strong evidence supported by reliable assessment or verification.

W_i is the criterion weight (%) and r_i the evidence score (0–5); all weights sum to 100.

$$S = \sum_i [\, W_i \times (r_i / 5) \,] \quad \text{where } 0 \le S \le 100$$

Example: scores 4, 4, 3, 4, 5 with weights 25, 20, 25, 15, 15 produce 78/100. Once rubric scores are set, calculation is deterministic.

IMPORTANT LIMIT A fixed formula prevents word count from entering the score directly, but it does not by itself stop a model from assigning higher evidence levels to verbose wording. Evidence extraction must also be tested and calibrated.

5.2 Evidence-based skill extraction

For each skill, record its name, source excerpt/location, evidence type, rubric level, uncertainty and verification status. “Implemented a Sudoku solver using backtracking in Java” supports a project claim; “Skills: Java” is only a claim; an assessment shows performance on that test. No single item proves universal mastery.

5.3 Capped keyword contribution

Preferred rule: raw keyword frequency earns no direct points. Keywords may retrieve evidence, not prove skill. If retained in a legacy feature, cap the frequency signal:

$$f(k) = \min(k / m, 1)$$

Here k is occurrence count and m the cap threshold. With m=2, counts 0, 1, 2, 5 and 10 yield 0, 0.5, 1, 1 and 1. Capping repetition does not prevent unsupported lists of different skills; evidence rules are still required.

5.4 Duplicate-evidence handling

1. Extract evidence statements and source locations.
2. Find exact duplicates and paraphrases of the same achievement.

3. Link repeats to one underlying evidence item.
4. Score unique evidence and enforce each criterion's maximum.
5. Count distinct achievements only where the rubric permits.

5.5 Robustness property

For qualification-preserving transformation T: S(R) approximately equals S(T(R))

R is the original resume; T(R) is a transformed version with the same job-relevant evidence. With identical extracted evidence and deterministic scoring, scores should match. Extraction errors may still cause variation, so test this property empirically.

6. Experimental Design and Evaluation Metrics

Validate observed behavior, not just the formula, by comparing the baseline and proposed pipeline on the same resumes and job descriptions.

6.1 Test dataset and controlled transformations

Start with at least 50 base resumes across three job descriptions, subject to consent, data quality and access. This is an engineering pilot, not a statistically powered guarantee; expand before making broad claims.

Variant	Controlled change	Expected result
Original	Use the base resume.	Record baseline score and rank.
Concise	Remove redundancy while preserving every relevant qualification.	No material score drop.
Longer / irrelevant	Add text that does not add job-relevant evidence.	No material score gain.
Keyword-stuffed	Repeat existing skill terms without new evidence.	No material score gain.
Paraphrased	Express the same project or skill evidence in different words.	Similar score if meaning is preserved.
Duplicate evidence	Repeat the same achievement in multiple sections.	No extra credit for the same achievement.
New valid evidence	Add a genuinely relevant, supportable achievement.	A score change may be appropriate.

Check that transformations preserve qualifications. Keep rubric, model, prompts and scoring settings consistent across comparisons; repeat selected runs to estimate variability.

6.2 Metric A - Mean score sensitivity

MSS = (1 / N) x sum_i |S_i - S_i'|

Compare each original score S_i with transformed score S_i'. Lower mean absolute change indicates greater stability. Also report maximum change and results by transformation to expose outliers.

6.3 Metric B - Keyword-stuffing gain

KG = S(R_stuffed) - S(R_original)

Without new evidence, the desired gain is approximately zero. Report average and worst-case gain. A rise from 65 to 84 caused only by repeated keywords is a failure.

6.4 Metric C - Shortlist stability

$$J(A, B) = |A \cap B| / |A \cup B|$$

A is the original top-K shortlist and B the transformed shortlist. Jaccard=1 means identical shortlists. Inspect cutoff cases, where small score changes can alter selection.

6.5 Metric D - Qualification precision and recall

$$\text{Precision} = TP / (TP + FP) \qquad \text{Recall} = TP / (TP + FN)$$

TP = qualified candidates correctly shortlisted; FP = shortlisted candidates who do not meet the standard; FN = qualified candidates excluded. Use an agreed rubric and independent reviewers to establish reference labels; an AI score is not ground truth.

6.6 Metric E - Group-level error analysis

Where lawful and privacy-protective, compare selection and false-negative rates across relevant groups. Use group data for auditing, not individual scoring. Length robustness does not guarantee demographic fairness; investigate material disparities [1, 6].

6.7 Proposed initial pilot criteria

Metric	Proposed starting target	Interpretation
Mean absolute score change	<= 2 points out of 100	For qualification-preserving transformations.
Mean keyword-stuffing gain	<= 1 point	Adding repeated keywords should have little or no effect.
Top-K Jaccard similarity	>= 0.90	Shortlist membership should remain stable.
Precision / recall	Set a benchmark before the pilot	Targets depend on role and qualification labels.
Group-level disparities	Investigate material differences	Use appropriate data, methods and privacy controls.

TARGETS ARE NOT RESULTS These are proposed engineering thresholds, not universal fairness standards or measured Niyukti results. Final targets should be justified against baseline performance, error costs and validation evidence.

8. Limitations, Risks and Human Oversight

Risk	Why it matters	Control
Incomplete resume evidence	A concise resume may omit a skill the candidate genuinely has.	Treat missing evidence as unestablished, not proof of inability; allow assessment or clarification.

Risk	Why it matters	Control
LLM extraction mistakes	The model may misread a project or infer unsupported skills.	Retain source excerpts, use structured output and route uncertain cases to review.
Assessment bias	Tests may favor particular backgrounds, time availability or resource access.	Validate test design, accessibility, scoring consistency and role relevance.
Duplicate / paraphrased claims	The same achievement may be presented several ways.	Link claims to unique underlying evidence and limit criterion contributions.
Human reviewer bias	Human review does not automatically remove bias.	Use a shared rubric, record override reasons and audit decisions.
Privacy and sensitive data	Resume data and group-level audits can expose sensitive information.	Minimize data, restrict access, set retention rules and use group data only where lawful and appropriate.

9. Conclusion and Expected Outcomes

Team Atom proposes fixed role-specific scoring, evidence extraction, duplicate control and controlled resume-variant testing to mitigate length and keyword bias.

Test the core claim: when qualifications are unchanged, irrelevant length or keyword changes should not materially alter scores or shortlist outcomes. Measure score sensitivity, keyword gain, top-K Jaccard similarity, precision, recall and group-level disparities.

The intended benefit is more consistent, explainable shortlisting. It remains a hypothesis until tested. Report results and limitations, and revise failures; the goal is to measure, reduce and monitor specific risks—not promise zero bias.10. References and Appendix

10.1 References

[1] Wilson, K. & Caliskan, A. (2024). “Gender, Race, and Intersectional Bias in Resume Screening via Language Model Retrieval.” AIES, 7(1), 1578–1590. https://doi.org/10.1609/aies.v7i1.31748. Reports demographic and document-length effects in resume retrieval.

[2] scikit-learn. “TfidfVectorizer” documentation. https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html. TF-IDF weighting and normalization options.

[3] Robertson, S. & Zaragoza, H. (2009). “The Probabilistic Relevance Framework: BM25 and Beyond.” Foundations and Trends in Information Retrieval, 3(4), 333–389. https://doi.org/10.1561/1500000019.

[4] Sentence Transformers. “Semantic Search” documentation. https://www.sbert.net/examples/sentence_transformer/applications/semantic-search/README.html.

[5] Varshney, A. & Ganuthula, V. R. (2025). “Signal or Noise? Evaluating Large Language Models in Resume Screening Across Contextual Variations and Human Expert Benchmarks.” arXiv:2507.08019. https://arxiv.org/abs/2507.08019. Preprint; interpret in light of its study design and status.

[6] NIST (2023). “Artificial Intelligence Risk Management Framework (AI RMF 1.0).” https://www.nist.gov/itl/ai-risk-management-framework. Guidance for managing AI trustworthiness and risk.